

Joseph Barbosa

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EDUCATION

University of Waterloo (Double Degree with Wilfrid Laurier University) GPA: 3.98 / 4.0 (92%)
Bachelor of Computer Science (AI Specialization, Statistics Minor) & Bachelor of Business Administration Sep. 2025 – Present

EXPERIENCE

Undergraduate Research Assistant Incoming Sep. 2026
University of Waterloo Waterloo, ON

- Evaluating ASR models on a **380-hour** code-switched Kazakh-Russian medical transcript corpus under Professor Helen Chen; analyzing transcription errors under language switching to improve robustness for AI-assisted clinical documentation.

Software Engineering Intern May 2026 – Aug. 2026
Scotiabank Toronto, ON

- Spearheaded Flowpilot: a VLM-powered ReAct QA agent using Set-of-Marks grounding for live UI test automation. Achieved a **78%** one-shot pass rate on a held-out **33-case** eval suite. Reduced QA workflows from **2h to 20m** across **5 services**.
- Designed AI architecture for ProfileAI (shipping FY27, **50K clients/day**): vector search, pre-filtering, and prompt caching pipelines; collaborated with MongoDB engineers on production-scale inference and retrieval.

Case Team Delegate Jun. 2026 – Present
Wilfrid Laurier University Waterloo, ON

- Selected as **1 of 4** delegates for Laurier's international case team; competing in weekly time-bound case simulations (5–12h).
- Designed an AI-assisted team workflow (mockups, transcriptions, and solutioning), adopted across all weekly competitions.

Software Engineering Intern Jan. 2026 – Apr. 2026
ESGTree Waterloo, ON

- Built a production NL-to-SQL agent over a **98-table** MySQL database using ReAct reasoning, schema retrieval, and join-aware prompting; achieved **90.9%** accuracy on a **77-query** eval suite, crediting self-correction (**+13.0pp**) & RAG (**+7.8pp**) gains.
- Engineered production backend with SQL validation, read-only DB access, and multi-tenant isolation via JWT; designed a 3-model cost/latency tier and Docker-sandboxed Python execution for data analytics through a Next.js interface.

Software Development Intern Jun. 2024 – Dec. 2024
FuturIQ Brampton, ON

- Built an end-to-end ML pipeline for housing price prediction: automated Selenium data ingestion of **1,000+** current listings and engineered **8** features, deploying the regularized polynomial model via Flask/React with real-time auto-retraining.

PROJECTS

Scholr | *Python, FastAPI, sentence-transformers, MCP, Postgres* | [GitHub](https://github.com) tryscholr.com

- Built a multi-agent research assistant over **200M papers** using recursive planning and parallel retrieval. **300+ users**.
- Engineered two-stage retrieval (BGE-small, MiniLM reranking), improving nDCG@5 by **10.8%** & recall@5 by **6.3%**.
- Architected self-correction through citation validation & replanning; deployed via MCP and a Next.js web app.

Studeal | *Python, Chrome DevTools Protocol, asyncio, Celery, Postgres + pgvector* | [GitHub](https://github.com)

- Built an autonomous browser agent that searches online marketplaces from natural-language queries, navigating Kijiji, Facebook Marketplace, eBay, & Craigslist with zero site-specific code to extract, deduplicate, & rank live listings to user preferences.
- Engineered a scalable multi-agent system with context-isolated browser agents, asynchronous extraction workers, and a CDP-based perception layer, achieving **5/5** successful end-to-end evaluation runs while extracting **35-54** listings in **43-128s**.

PantryPal | *Python, PyTorch, QLoRA, DPO, XGBoost* | [GitHub](https://github.com) [Blog Post](#)

- Investigated whether weak domain classifiers could replace human preference labels for LLM alignment; trained **18** XGBoost reward models across dietary categories, generating **1,200+** DPO preference pairs.
- Fine-tuned Llama 3.2 3B with QLoRA SFT on **8K** recipes, reaching **99.7% compliance** vs. **56.8%** GPT-4o.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, MySQL, SQLite), TypeScript, JavaScript, HTML/CSS, C

AI / ML: Hugging Face Transformers, PEFT, LangGraph, LangChain, MCP, fine-tuning, RAG, AI evals, OpenAI API

Frameworks: React, Next.js, FastAPI, Flask, Node.js

Libraries: pandas, NumPy, scikit-learn, XGBoost, Pydantic, SQLAlchemy, Selenium, Playwright

Databases / Cloud: PostgreSQL, SQLite, Redis, pgvector, FAISS, GCP, Azure

Tools: Git, Docker, Bash, Claude Code, CI/CD, GitHub Actions

Awards: President's Gold Scholarship, Loblaw National Scholarship, Stanford ML Specialization, \$3,000+ in Case Winnings